

Traffic Demand Prediction System

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Project: INNOVEXA CATALYST Machine Learning Internship

Dataset: Metro Interstate Traffic Volume (48,000+ Records)

1. Data Cleaning

The foundation of the predictive model relied on a clean, consistent dataset. The initial preprocessing phase involved:

- **Missing Value Resolution:** Identified and removed null values specifically localized within the holiday column to ensure uninterrupted training sequences.
- **Deduplication:** Scanned the dataset for identical rows and successfully removed 17 duplicate entries, preventing artificial weighting of specific traffic events.

2. Exploratory Data Analysis (EDA)

To understand the underlying distribution of the traffic data, extensive visual analysis was conducted:

- **Distribution Analysis:** Countplots and histograms were utilized to map the overall distribution of the target variable (traffic_volume), identifying normal flow versus extreme congestion events.
- **Categorical Insights:** Barplots were deployed to analyze average traffic volumes across categorical variables like weather types and holidays.
- **Correlation Mapping:** Heatmaps were generated to establish initial bivariate relationships, immediately highlighting the strong connection between time of day and traffic density.

3. Feature Engineering

Raw data was transformed into highly predictive signals. A critical architectural decision was made to **strictly avoid creating a traffic_density_score**. Deriving a feature directly from the target variable (traffic_volume) causes target leakage, which invalidates the model in real-world scenarios. Instead, the following features were engineered:

- **Time-Based Cyclic Encoding:** Time is cyclical; 23:00 and 00:00 are chronologically adjacent but numerically distant. hour_sin and hour_cos features were engineered using trigonometric transformations to allow the models to comprehend the continuous loop of a 24-hour day.
- **Routine Flags:** Created binary indicators such as weekend_flag and rush_hour to explicitly flag periods historically prone to high or low volume.

4. Feature Encoding

Machine learning algorithms require numerical inputs. Categorical data was seamlessly translated:

- **Weather Conditions:** Text-based columns describing weather severity (e.g., "Clear", "Clouds", "Rain", "Snow") were mapped to distinct numerical identifiers, allowing the tree-based models to split nodes based on environmental impact.

5. Feature Selection

Not all engineered features contributed positively to the model. Feature selection was conducted using Correlation Heatmaps and Random Forest Feature Importance validation experiments.

- **Dimensionality Reduction:** Features demonstrating negligible predictive importance—specifically `snow_1h`, `holiday_new`, `weekend_flag`, and `rush_hour`—were removed.
- **Validation:** Stripping these redundant features maintained model performance (the reduced model achieved an R2 score of **0.9544** compared to **0.9545** for the full model) while successfully reducing structural complexity and noise.

6. Model Training

Following a strict 80/20 train-test split, the data was passed through progressively advanced algorithms:

- **Random Forest (Baseline):** Established a strong initial benchmark using bagging techniques.
- **XGBoost Regressor:** Optimized using `RandomizedSearchCV` (adjusting `max_depth`, `learning_rate`, and applying strict subsampling) to sequentially minimize residuals and handle structural edge cases.
- **LightGBM Regressor:** Tuned for hardware-efficient, leaf-wise growth, allowing the model to aggressively target granular data points (like precise temperature fluctuations) using histogram-based binning.
- **Weighted Ensemble:** The final architecture combined the two optimized boosters using the precise Innovexa Catalyst specification: **55% LightGBM + 45% XGBoost**.

7. Models Comparison

The performance of the models was evaluated against the holdout test set using R-squared (R2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

Model Architecture	R2 Score	RMSE	MAE
Random Forest (Baseline)	0.9544	426.89	239.37
LightGBM (Optimized)	0.9557	420.61	252.96
XGBoost (Optimized)	0.9576	411.53	246.68
Final Ensemble (55/45 Blend)	0.9576	411.43	245.59

Observation: The weighted ensemble successfully minimized the overall error rates (RMSE and MAE) by allowing the models to cancel out each other's specific blind spots.

8. Conclusion

The Metro Interstate Traffic Prediction project yielded a highly accurate, deployment-ready ensemble model. A critical insight extracted from the final feature importance analysis revealed the hierarchy of traffic drivers:

- **The Dominance of Time:** Temporal patterns (hour_cos, hour_sin, and dayofweek_new) accounted for over 93% of the total decision-making power of the baseline models.
- **The Role of Environment:** Traffic volume is primarily driven by human routines (daily commutes, weekend rest). Weather factors (temperature, rain, cloud cover) play a strictly secondary role, only heavily influencing the model during extreme anomalies.

The final deployed pipeline efficiently processes real-time user inputs, applies cyclic transformations, and leverages a 55/45 gradient boosting ensemble to forecast traffic demand with exceptional reliability.